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6CS012

Artificial Intelligence and Machine Learning

Portfolio Task 1: QnA

Full Name	:	Narayan Lohani
Student Number	:	2408869
Course	:	BSc (Hons) Computer Science
University Email	:	n.lohani@wlv.ac.uk
Group	:	L6CG15
Tutor	:	Jinu Nyachhyon
Module Leader	:	Siman Giri
Submission Date	:	May 10, 2026

Long Question Answer

1. You are a Data Scientist at a leading digital payment platform such as eSewa, working on large-scale user transaction data.

- Identify two high-impact business problems where unsupervised learning can provide value (e.g., customer segmentation, fraud detection, behavioral profiling).
- For each problem:
  - Clearly define the problem in a real-world context.
  - Propose a suitable unsupervised learning approach (e.g., clustering, anomaly detection, dimensionality reduction) and recommend specific algorithms (e.g., K-Means, DBSCAN, Autoencoders).
  - Discuss how the model outputs can be integrated into real-time or batch systems to support business decisions.
  - Briefly mention one limitation or challenge of your proposed approach.
- Assume limited labeled data and large-scale transactions.

Answer:

Problem Identification

Two high-impact problems include anomaly detection for fraud prevention and behavioral customer segmentation. These leverage unsupervised methods to process large-scale data, reducing fraud losses and enabling personalized services.

Fraud Anomaly Detection

In a real life context at digital payment systems, fraud detection involves scanning millions of daily transactions such as transfers, bill payments, or wallet loads. While scanning if there exists unusual patterns such as sudden high-value transfers to new recipients or atypical device location combinations that may be potential fraud attempts.

2 We can use Isolation Forest, an anomaly detection algorithm that isolates outliers by randomly partitioning data. This is ideal for large-scale, high-dimensional transaction logs as Isolation

Forest has linear time complexity and minimal hyperparameters which is suitable for unlabeled data by assuming anomalies require fewer splits to isolate.

Isolation Forest can be deployed in real-time systems via streaming pipelines such as Apache Kafka. When the scores go above a threshold, it alerts for transaction holds or user verification. New thresholds are adapted on a fixed period (usually at night when traffic is low) by retraining on a new batch of aggregated data.

The most challenging part of using Isolation Forest for Fraud Detech is the model will get worse at detecting frauds when the fraudsters change their scamming method. To fix this, we have to regularly update the AI using recent data, even if we don't have the final ‘answers' yet on which new transactions were definitely scams.

Customer Behavioral Segmentation

One way to tailor offers like cashback or payment limits is by profiling users based on unlabeled transaction data. This is done by distinguishing high-frequency small payments (daily expenses) and low-frequency large payments (bulk transactions).

Therefore, we can apply DBSCAN clustering, which groups variable-sized clusters based on density without predefined cluster counts. DBSCAN clustering performs very well for large-scale heterogeneous data by handling noise and outliers effectively, unlike K-Means.

To implement DBSCAN clustering, every week a big update is run on past week data which then categorizes users to find “High-Spender", "Frequent Small Shopper","Potential Risk" types of users. These data are sent to marketing systems to send targeted ads, or to set spending limits for users who look risky.

The biggest challenge of DBSCAN is avoiding the curse of dimensionality. The raw data contains high-dimensional features like transaction timestamps and amounts. PCA, Latent Space of AutoEncoder can be used to reduce the dimensions.

Short Question Answer

Overfitting

1. In the context of machine learning:

- Define overfitting and underfitting and clearly distinguish between the two in terms of model behavior on training and unseen data.
- Explain why both overfitting and underfitting negatively impact generalization performance.
- Illustrate your explanation using simple examples (e.g., a model memorizing training data vs failing to capture complex patterns in the data).

Answer:

Underfitting occurs when a model fails to learn the training data and does not capture the underlying patterns yielding high train error and high-test error as the model does not have enough complexity.

3 Overfitting occurs when a model learns the training data so precisely that it even learns the noise yielding low train error but high-test error because model cannot generalize correctly.

5 Both result in poor generalization performance. Underfitting cannot capture the patterns of
6 the train data resulting in both high train loss and test loss (high bias) and Overfitting
4 memorizes noise, causing a high difference between train loss and test loss (high variance).As neither has a balance between bias and variance, generalization is poor.

An example of underfitting can be a linear model fitting quadratic data. A linear model does not have enough complexity to capture the curvature pattern of the training data as well as test data.

An example of overfitting can be a high-degree polynomial regression model learning the noisy training data perfectly but failing to converge on test data as it memorized the fluctuations rather than trend of the data.

Neural Network Architecture

2. FCN vs. Autoencoder:

- Compare a standard feedforward neural network (fully connected network) with an autoencoder in terms of:
  - Architecture structure
  - Objective/functionality
  - Input-output relationship
- Provide a simple example illustrating how an autoencoder differs from a standard neural network.
- Describe at least one application of autoencoders (e.g., dimensionality reduction, anomaly detection, image denoising).
- Explain how autoencoders help solve the chosen problem.

Answer

	Feedforward Neural Network	**Autoencoder**
Architecture	Input is fully connected to the hidden layer. Every hidden layer is connected to every forward hidden layer. The penultimate (last hidden) layer is connected to the output layer.	Multiple Convolutional layers with decreasing feature map size (Encoder) are connected compressing to bottleneck. Multiple Transposed Convolutional layers with increasing feature map (Decoder). It is expected to be symmetric with latent space in the middle.
Objective	Supervised learning for classification or regression (e.g., minimize cross-entropy loss).	Unsupervised reconstruction of input (e.g., minimize MSE between input/output).
Input-Output relationships	Input features + labels for outputs predictions.	Input only (self-supervised) for outputs reconstructed input.

For an image of handwritten digit “0” in Mnist dataset, an autoencoder will be used to reconstruct a noisy image of “0” to clean image of “0”, but a standard neural network will be used to classify the image as "0”.

An application of autoencoders is anomaly detection (e.g., fraud detection in credit card transactions). The latent space of autoencoders forces dimensionality reduction which captures normal patterns in the latent space. However, upon reconstruction, error is low for normal transactions but high for anomalies.

References

Haddad, M. F. C., 2021. Harnessing the power of intersection for pattern recognition: a novel unsupervised learning method and its application to financial engineering. *Engineering Reports,* 3(4), pp. 1-31.